Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/06/2005
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Project Title:	Effect Of Gravity On DNA Transactions In Mammalian Cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	29208	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Waldman, Barbara Criscuolo (University of South Carolina)		
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	Genetic recombination plays fundamental roles in many biological processes including regulation of gene expression, DNA repair, disease etiology, and evolution. For over a decade, we have used an experimental system to study intrachromosomal recombination in mammals. We introduce DNA substrates containing two different defective herpes simplex virus thymidine kinase (tk) gene sequences into mouse L cells that are tk-deficient. The pair of defective tk sequences resides as a closely linked direct repeat in the genome. Cells lines are expanded and plated into selective medium in order to recover tk-positive segregants that had undergone a recombination event. From the number of tk-positive segregants that arise, we calculate the rate of recombination in terms of number of events per cell per generation. We also analyze recombination products at the nucleotide level. Recently, we have devised a scheme for studying DNA double-strand break-induced recombination. We now propose to us our model system to study the effects of varying gravitational force on spontaneous and damage-induced recombination in mammals. We will use the Hypergravity Facility for Cell Culture to examine the effects of hypergravity of 4g and we will use a Rotating Cell		

Task Description:	Culture System to study the effects of simulated microgravity. The following questions will be addressed: 1.What influence does altered gravitational force have on the rate of intrachromosomal homologous recombination and the relative proportions of reciprocal versus nonreciprocal genetic exchanges in mammalian chromosomes? 2.What influence does altered gravitational force have on the types of mutations that may be induced at DSBs in mammalian chromosomes? 3.What influence does altered gravitational force have on the sensitivity of intrachromosomal recombination in mammalian chromosomes to different degrees and lengths of sequence identity? 4.What influence does altered gravitational force have on the amount of genetic information exchanged during recombination in mammalian chromosomes? We hope to better our understanding of the role of the genome in sensing and responding to gravitational force, to increase our understanding of the possible role of gravity in evolution, and to gain important knowledge relevant to the risks faced by humans in the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	To this point, we have conducted three experiments using the Hypergravity Facility for Cell Culture (HyFaCC) at the Ames Research Center (ARC) in Palo Alto. For each experiment, cells containing an integrated recombination substrate were shipped to ARC and grown for 4 days on the HyFaCC at 4g or in a control incubator at 1g. The cells were then shipped back to our lab at the University of South Carolina for recombination rate determination and analysis of recombinants. An additional set of cells remained in South Carolina and served as a control for studying recombination in cells that were not exposed to the stresses of shipping. The goal of this entire experiment was to assess the effect of hypergravity on the rate and nature of chromosomal recombination in mammalian cells. Based on our three experiments, we have not obtained statistically significant evidence that growth of cells under conditions of hypergravity influences recombination. However, our results suggest a trend in which hypergravity may be associated with an increase in intrachromosomal recombination rate. The average rate of recombination for the stationary control cells for all three experiments was 8.88 X 10-7 (events per cell per generation), while average rates of recombination for the cells sent to ARC and grown at 1g and 4g were 11.0 X 10-7 and 15.75 X 10-7, respectively. For the stationary control cells, out of 97 recombinants analyzed, 91 recombinants were gene conversions and 6 were crossover events. For the cells grown at 1g at ARC, out of 102 recombinants analyzed, 93 recombinants were gene conversions while nine were crossover events. For the cells grown at 4g at ARC, out of 92 recombinants analyzed, 84 recombinants were gene conversion while eight were crossover events. We have also initiated experiments to study the impact of simulated microgravity on recombination, using a Rotary Cell Culture System. Thus far, we have obtained no evidence that microgravity affects spontaneous homologous recombination in mammalian chromosomes.
Bibliography Type:	Description: (Last Updated: 03/20/2017)
Abstracts for Journals and Proceedings	Waldman AS, Bhattacharjee V, Waldman BC. "Effect of hypergravity on intrachromosomal homologous recombination in mammalian cells." American Society for Biochemistry and Molecular Biology: 2004 Annual Meeting, Boston, MA, June 12-16, 2004. FASEB Journal. 2004 May;18 Suppl:Abstract C87. , May-2004